

# Benjamin P. Zuk

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## EDUCATION:

**Villanova University**  
**Master of Science: Mechanical Engineering**  
Cumulative GPA: 3.4/4.00

**Villanova, PA**  
**Expected September 2013**

**The Pennsylvania State University**  
**Bachelor of Science: Mechanical Engineering**  
Cumulative GPA: 3.21/4.00

**University Park, PA**  
**Class of May 2011**

## EXPERIENCE:

### **Laboratory for Advanced Thermal and Fluid Studies**

#### *Thesis Research*

- Designed and built an evaporative flow loop to cool high power electronics
- Built a MATLAB model to specify design geometry for custom cold plates
- Tested custom designed heat sinks in high power electronics to validate design

**Villanova, PA**  
**January 2012-Present**

### **Siemens Healthcare Diagnostics**

#### *Reliability Engineer*

- In charge of writing failure analysis software for a new instrument platform
- Determined failure modes of components of the Dimension Vista assay instrument
- Built testing equipment and tested components to reproduce these failures

**Glasgow, DE**  
**May 2011- Dec 2011**

### **OpSec Security**

#### *Process Engineer*

- Developed security products for automobile company in Germany
- Developed lamination and printing procedures for the new Belize passport
- Wrote company barcoding standard that allowed for greater production efficiency

**Lancaster, PA**  
**June 2010 – August 2010**

### **Formula SAE Team**

#### *Powertrain & Composites Leader*

- Lead engineer in charge of powertrain, turbocharger, and composite systems
- Developed testing system to analyze fluid flow through intake and exhaust systems
- Built a data acquisition system to analyze powertrain system

**University Park, PA**  
**August 2007-May 2011**

### **The Learning Factory- Penn State University**

#### *Teaching Assistant in the Mechanical Engineering Machine Shop*

- Oversaw undergraduate capstone and graduate research projects
- Taught weekly classes in manual mill, manual lathe, MIG and TIG welding
- Gained experience in rapid prototyping and CNC machining methods

**University Park, PA**  
**2009-2011**

## RELEVANT COURSES:

Thermodynamics, Measurement and Instrumentation, Heat Transfer, Modeling Dynamic Systems, Rocket Propulsion, Compressible Flow, Advanced Fluid Mechanics, Convection Heat Transfer, Micro-Scale Heat Transfer, Two Phase Flow, Principles of Turbo Machinery

## COMPUTER SKILLS:

MATLAB, C++, Arduino C, Solidworks, Pro Engineer, CATIA, ANSYS Workbench, ANSYS FLUENT, ANSYS CFX, OMAX Suite, National Instruments LabVIEW, CatalystEX, COMSOL, Mathcad